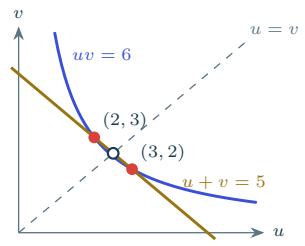
ALGEBRA ON ONE PAGE

Solving a quadratic as a line meeting a hyperbola

Factorising asks for two numbers with a given sum and product, which is where the line $u+v=s$ meets the hyperbola $uv=P$.

1 / THE RULE AND THE PICTURE

Factorising ax^2+bx+c as $a(x-p)(x-q)$ needs $p+q=-\frac{b}{a}$ and $pq=\frac{c}{a}$



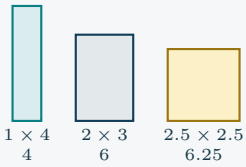
Expanding $(x-p)(x-q)=x^2-(p+q)x+pq$ and matching with $x^2+\frac{b}{a}x+\frac{c}{a}$ turns the quadratic into a pair of simultaneous equations, one linear and one reciprocal:

$$\boxed{u+v=s=-\tfrac{b}{a}, \qquad uv=P=\tfrac{c}{a}}$$

The roots are the coordinates of the points where the line meets the hyperbola, and since swapping u and v changes neither equation the points come in mirror pairs (p,q) and (q,p) across the diagonal $u=v$. For x^2-5x+6 they are $(2,3)$ and $(3,2)$, with the midpoint $(2.5,2.5)$ on the diagonal.

2 / THE AREA PICTURE / SUM AND PRODUCT OF A RECTANGLE

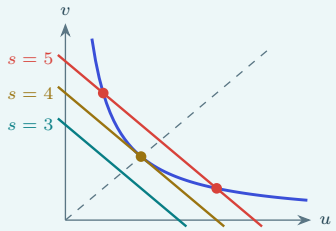
A rectangle with sides p and q has semi-perimeter s and area P, and the square has the most area



With $s=5$ the rectangles 1×4, 2×3 and 2.5×2.5 have areas 4, 6 and 6.25, so no rectangle with semi-perimeter 5 can have area above $6.25=(s/2)^2$, which is the arithmetic and geometric mean inequality. Real roots therefore exist exactly when $P\le s^2/4$, that is $s^2-4P\ge0$, the discriminant b^2-4ac once $a=1$; x^2-5x+7 would need an area of 7 and has none.

3 / SLIDING THE LINE / THE DISCRIMINANT AS A PICTURE

On $uv=4$ the lines $u+v=5,\ 4,\ 3$ cross twice, touch once and miss

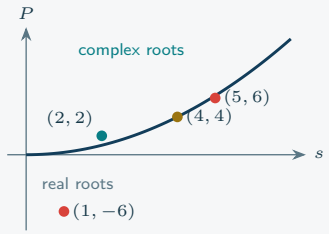


s	quadratic	Δ	picture
5	$x^2-5x+4=(x-1)(x-4)$	9	crossings $(1,4),(4,1)$
4	$x^2-4x+4=(x-2)^2$	0	tangent at $(2,2)$
3	x^2-3x+4	-7	line passes below

The tangent line touches at the square $(2,2)$ on the diagonal. For $s=3$ the midpoint $(1.5,1.5)$ is nearer the origin than the hyperbola's vertex, so the roots $\frac{3\pm i\sqrt{7}}{2}$ are complex.

4 / THE (s,P) PLANE / EVERY QUADRATIC IS A POINT

The parabola $P=s^2/4$ separates real roots below it from complex roots above it



The point $(5,6)$ is x^2-5x+6 with real roots, $(4,4)$ is the double root on the boundary, and $(2,2)$ is x^2-2x+2 with roots $1\pm i$. A negative product, such as $(1,-6)$ for $x^2-x-6=(x-3)(x+2)$, puts the hyperbola in the second and fourth quadrants where the line always crosses it, so the roots are real with opposite signs, at $(3,-2)$ and $(-2,3)$.

5 / THE FORMULA AS A PICTURE / MIDPOINT PLUS OR MINUS THE HALF GAP

The roots sit symmetrically about $s/2=-\frac{b}{2a}$, the vertex of the parabola

$$p,\ q=\frac{s}{2}\pm\sqrt{\frac{s^2}{4}-P}=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

For $2x^2-10x+12=0$ divide by 2 to get $s=5$ and $P=6$, so the roots are 2 and 3. For $x^2-4x+1=0$ we have $s=4$, $P=1$, midpoint 2 and half gap $\sqrt{4-1}=\sqrt{3}$, so the roots are $2\pm\sqrt{3}\approx3.73,\ 0.27$, the points $(2+\sqrt{3},\ 2-\sqrt{3})$ and its mirror image.

ASIDE / WHY SYMMETRIC

The pair (p,q) is unordered, so the equations can only ask for $p+q$ and pq, the two quantities that do not change when p and q swap.

ASIDE / CHECK IN PYTHON

`np.roots([1,-4,1])` gives $3.732,\ 0.268$, and `np.roots([1,-3,4])` gives $1.5\pm1.32j$.

Convention. $s=-b/a$ and $P=c/a$; real roots need $s^2\ge4P$. After the earlier sheets on lines meeting curves.

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